



**Irkutsk National Research Technical University**  
**Baikalsk school of BRICS**

09.04.02 Information systems and technology  
Program: Information technologies, networks and big data

# Course: Data analysis

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## Statistical Analysis

- Descriptive Statistics,
- Inferential Statistics (Statistical Tests),
- Exploratory data analysis (EDA)

# Data Sets



**Name:** “data\_for\_analysis“ (from data frames factor\_df and imputed\_handle\_MD\_df\_final q.v. practice 1)

**Source:** Internal company database

**Size:** 1148 obs. of 31 variables

**Format:** CSV

## Columns:

record\_id (int) Unique order identifier

outcome (int) Presence of a tumor

hormone1 (num) result of hormone 1 level assessment

...

hormone4 (num) result of hormone 4 level assessment

h\_index\_34 (num) index (hormonal ratio)

...

...

antioxidant5 (num) result of antioxidant 5 level assessment

factor\_eth (int) ) Ethnicity

factor\_h (int) Increased hormones

factor\_pcos (int) Diagnosis

factor\_prl (int) Increased hormone

## Potential Use for the Practical class 3:

select the most optimal distribution law and evaluate its parameters;

creating a table with descriptive statistics for a data set;

# Descriptive Statistics



```
install.packages("gtsummary")
```

```
library(gtsummary)
```

```
tbl_summary(data_for_analysis) # Automatic table
```

```
tbl_summary(data_for_analysis, by = outcome) # By groups
```

Homework!

| Characteristic | N = 1,148 <sup>†</sup> |
|----------------|------------------------|
| record_id      | 1,258 (654, 1,946)     |
| outcome        | 160 (14%)              |
| Unknown        | 1                      |
| factor_eth     |                        |
| 1              | 725 (63%)              |
| 2              | 316 (28%)              |
| 3              | 106 (9.2%)             |
| Unknown        | 1                      |

| Characteristic | 0<br>N = 987 <sup>†</sup> | 1<br>N = 160 <sup>†</sup> |
|----------------|---------------------------|---------------------------|
| record_id      | 1,299 (681, 1,975)        | 943 (523, 1,646)          |
| factor_eth     |                           |                           |
| 1              | 636 (64%)                 | 89 (56%)                  |
| 2              | 258 (26%)                 | 58 (36%)                  |
| 3              | 93 (9.4%)                 | 13 (8.1%)                 |
| factor_h       | 197 (20%)                 | 23 (14%)                  |
| Unknown        | 12                        | 1                         |

# Inferential Statistics (Statistical Tests)

## B. Hypothesis Testing

Tests whether an observed effect is statistically significant.

### Steps:

#### State Hypotheses:

H0 (Null): "No effect" (e.g., drug has no effect).

H1 (Alternative): "Effect exists" (e.g., drug works).

**Choose Significance Level ( $\alpha$ ):** Typically 0.05 (5%).

#### Calculate Test Statistic:

**Z-test** (known  $\sigma$ ): 
$$z = \frac{\bar{x} - \mu}{\sigma / \sqrt{n}}$$

**t-test** (unknown  $\sigma$ ): 
$$t = \frac{\bar{x} - \mu}{s / \sqrt{n}}$$

#### Compare to Critical Value:

If  $|z| > 1.96$  (for  $\alpha = 0.05$ ), reject H0.

#### Conclusion:

**p-value**  $< \alpha$ : Reject H0 (significant result).

**p-value**  $\geq \alpha$ : Fail to reject H0.

# Inferential Statistics (Statistical Tests)

## C. Common Statistical Tests

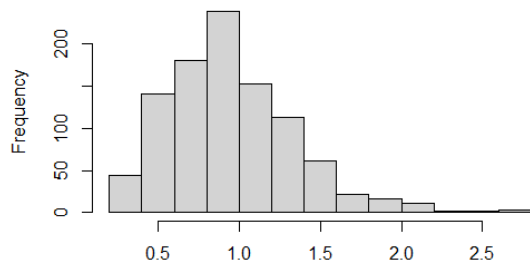
| Test               | Use Case                                  | Example                                |
|--------------------|-------------------------------------------|----------------------------------------|
| Independent t-test | Compare means of two groups.              | Men vs. women's salaries.              |
| Paired t-test      | Compare means of same group at two times. | Before vs. after a training program.   |
| ANOVA              | Compare means of three+ groups.           | Test scores across 3 teaching methods. |
| Chi-square test    | Test categorical data relationships.      | Smoking vs. lung cancer incidence.     |

| Assumptions                                                                               | Recommended test                                | Description                                                                                                                                                                                           |
|-------------------------------------------------------------------------------------------|-------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1. Normal distribution within each group<br>2. Homogeneity of variances (equal variances) | Student's t-test                                | Parametric test comparing means. Requires normality and equal variances.                                                                                                                              |
| 1. Non-normal distribution<br>2. Presence of outliers<br>3. Small sample size             | Mann–Whitney U test (Wilcoxon rank-sum test)    | Non-parametric analogue of the t-test. Compares medians or distributions. No normality required.                                                                                                      |
| 1. Non-normal distribution<br>2. Heterogeneity of variances (unequal variances)           | Brunner–Munzel test (Generalized Wilcoxon test) | Non-parametric test of stochastic equality $P(X>Y)=P(Y>X)$ . Does not require equal variances or a location-shift model. More robust alternative to the Mann-Whitney test when variances are unequal. |

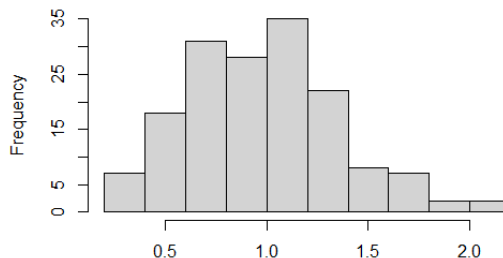
# Statistical Tests



Histogram of value\_outcome0



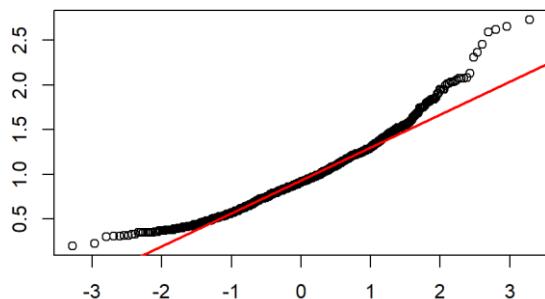
Histogram of value\_outcome1



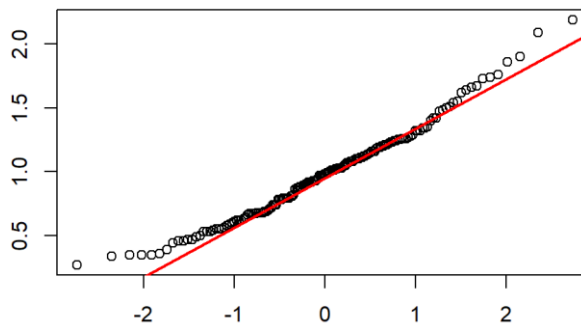
$N_1=160$

$N_2=987$

$p < 0.05$  confirms different variances



Shapiro-wilk normality test  
data: value\_outcome1 w =  
0.97772, p-value = 0.01093



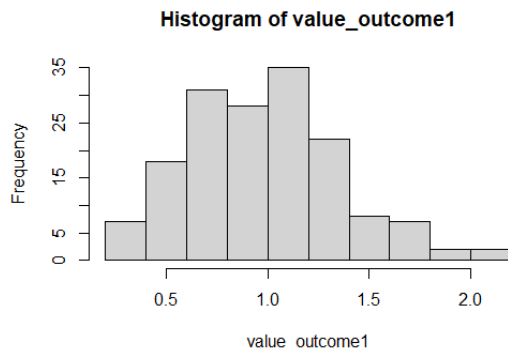
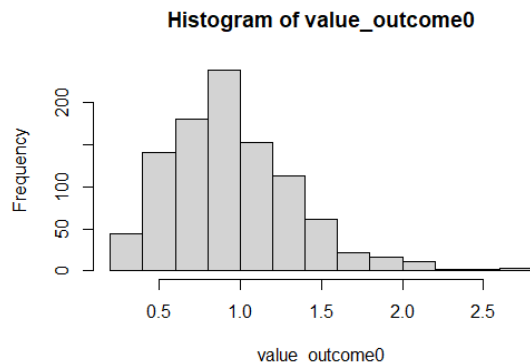
Shapiro-wilk normality test data:  
value\_outcome1 w = 0.97772, p-value =  
0.01093

Levene's Test for Homogeneity  
of Variance (center = median)  
Df F value Pr(>F) group 1  
0.0204 0.8866 1145

The p-value (0.01093) is less than the conventional significance level of 0.05. Therefore, we **reject the null hypothesis of normality**.



# Statistical Tests



$N_1=160$

$N_2=987$

The **Brunner-Munzel test** is ideal for comparing two groups with:

Unequal sizes

Different variances

Non-normal distribution

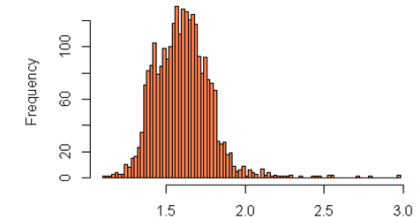
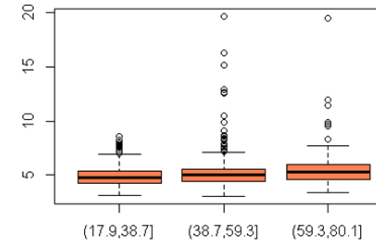
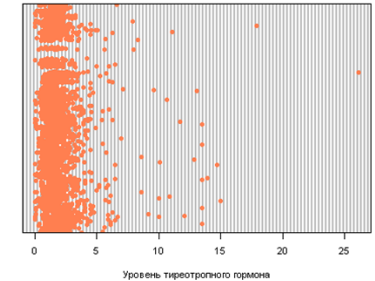
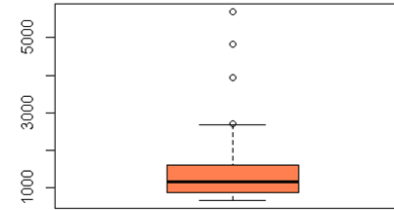
```
> car::leveneTest(lipids1 ~ outcome, data = data_for_analysis)
Levene's Test for Homogeneity of Variance (center = median)
      Df F value Pr(>F)      p < 0.05 confirms different variances
group   1  0.0204 0.8866
      1145
> brunner.munzel.test(group1, group2)

Brunner-Munzel Test

data: group1 and group2
Brunner-Munzel Test Statistic = 1.239, df = 214.4, p-value = 0.2167
95 percent confidence interval:
 0.4819959 0.5789400
sample estimates:
P(X<Y)+.5*P(X=Y)
 0.530468
```

# Exploratory data analysis (EDA)

|   |                                                                        |                                                            |
|---|------------------------------------------------------------------------|------------------------------------------------------------|
| 1 | Formulate biological hypothesis<br>Carry out experiment & collect data |                                                            |
|   | <b>Data exploration</b>                                                |                                                            |
| 2 | <b>1. Outliers Y &amp; X</b>                                           | <i>boxplot &amp; Cleveland dotplot</i>                     |
|   | <b>2. Homogeneity Y</b>                                                | <i>conditional boxplot</i>                                 |
|   | <b>3. Normality Y</b>                                                  | <i>histogram or QQ-plot</i>                                |
|   | <b>4. Zero trouble Y</b>                                               | <i>frequency plot or corrgram</i>                          |
|   | <b>5. Collinearity X</b>                                               | <i>VIF &amp; scatterplots<br/>correlations &amp; PCA</i>   |
|   | <b>6. Relationships Y &amp; X</b>                                      | <i>(multi-panel) scatterplots<br/>conditional boxplots</i> |
|   | <b>7. Interactions</b>                                                 | <i>coplots</i>                                             |
|   | <b>8. Independence Y</b>                                               | <i>ACF &amp; variogram<br/>plot Y versus time/space</i>    |
| 3 | Apply statistical model                                                |                                                            |



Zuur, Alain F., Elena N. Ieno, and Chris S. Elphick. "A protocol for data exploration to avoid common statistical problems." *Methods in ecology and evolution* 1.1 (2010): 3-14.

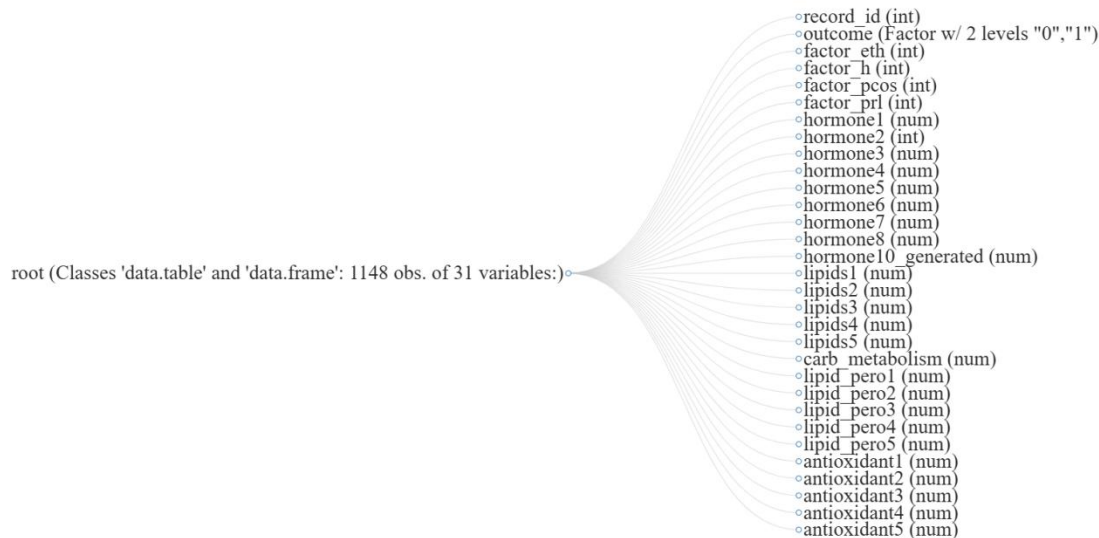
# Exploratory data analysis



library(DataExplorer)

## EDA Report

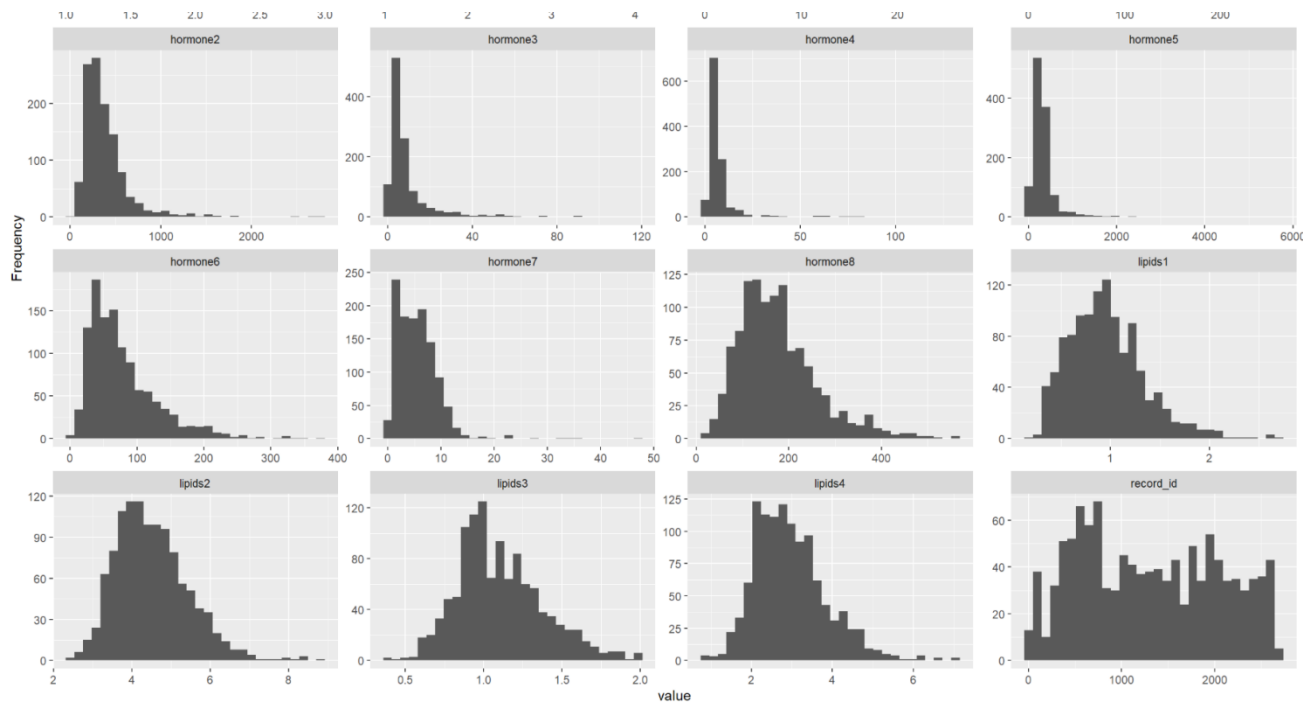
- Basic Statistics
  - Raw Counts
  - Percentages
- Data Structure
- Missing Data Profile
- Univariate Distribution
  - Histogram
  - Bar Chart (with frequency)
  - QQ Plot
- Correlation Analysis
- Principal Component Analysis



# Exploratory data analysis



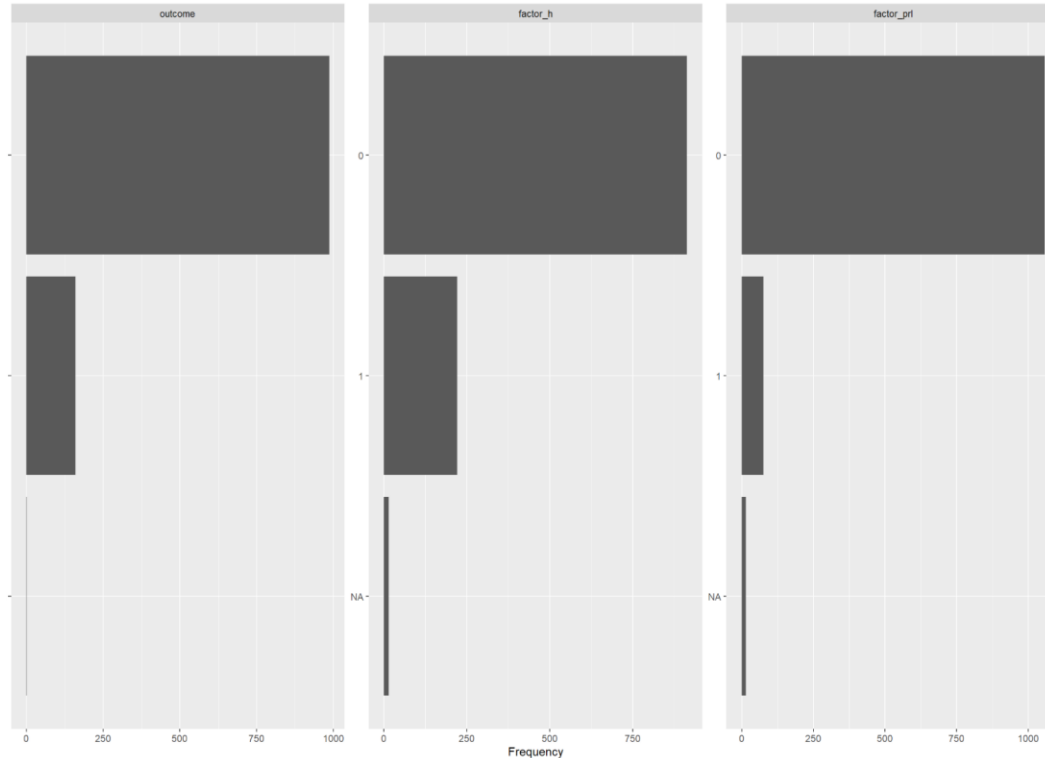
## Univariate Distribution



# Exploratory data analysis



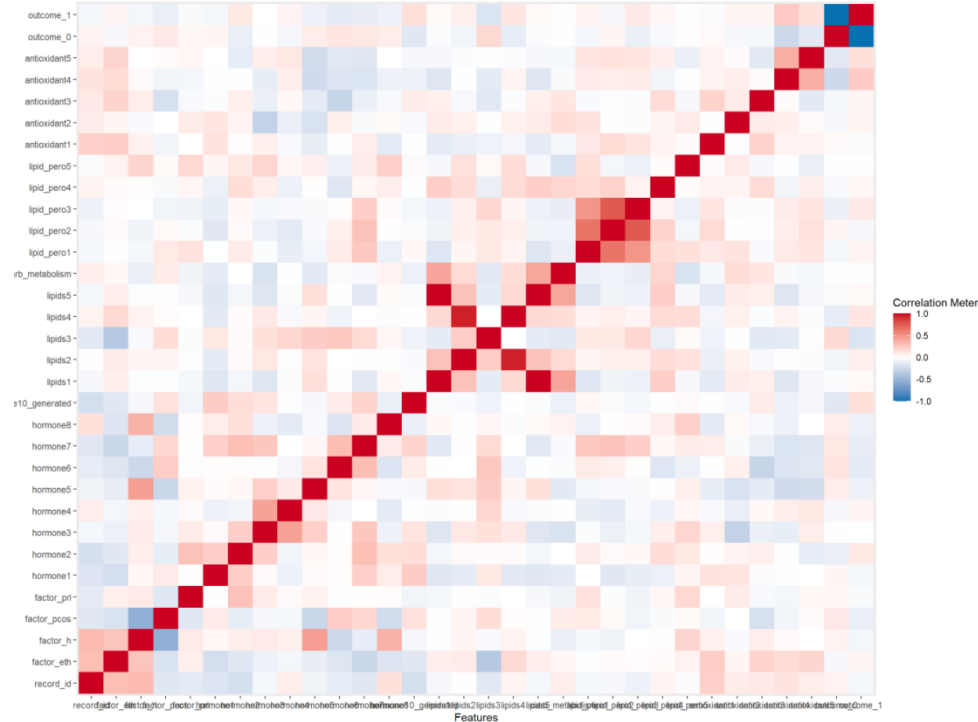
## Bar Chart (with frequency)



# Exploratory data analysis

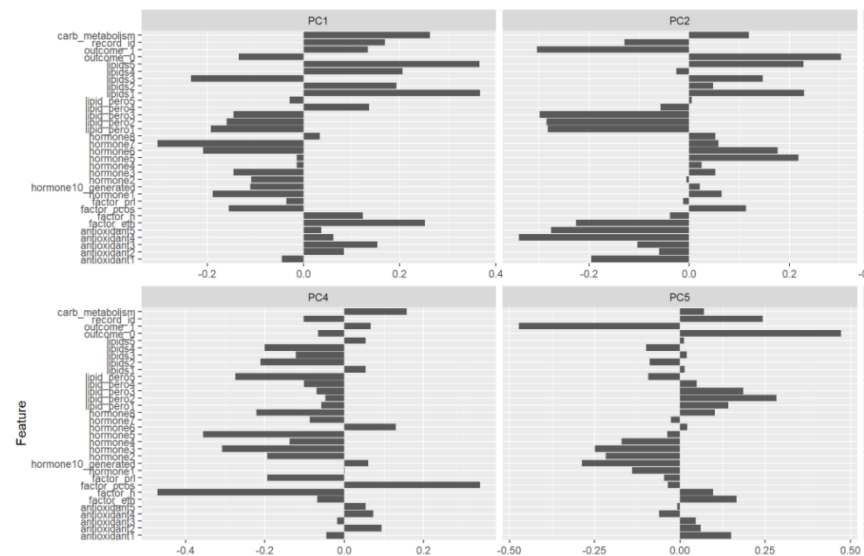
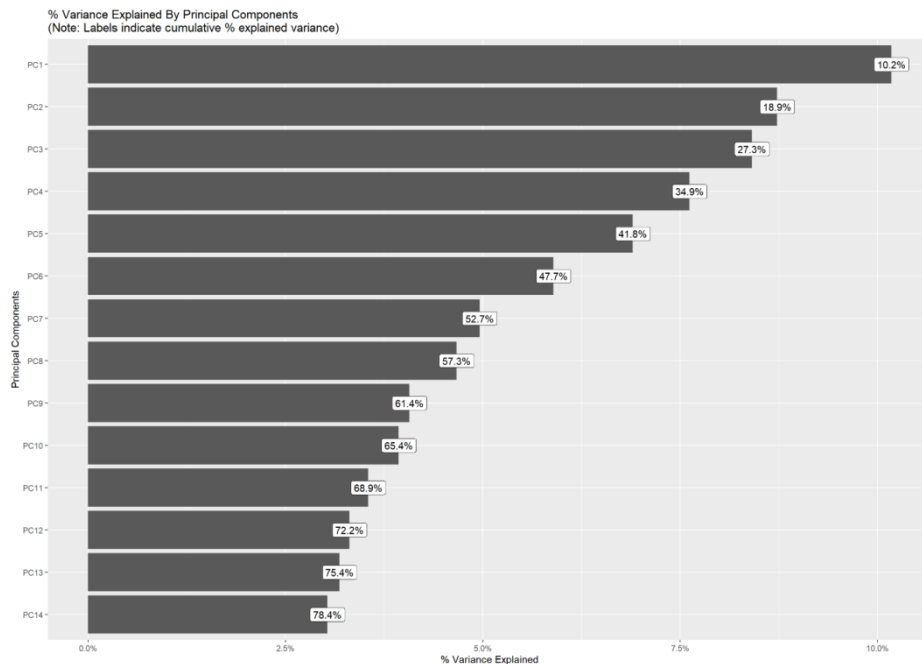


## Correlation Analysis





# Principal Component Analysis



# Task 3

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Deadline subgroup 1 20.04.2026 23:59

Deadline subgroup 2 27.04.2026 23:59)

## **Dataset “data\_for\_analysis”**

1. create a table with descriptive statistics and specify the parameters according to the selected distribution (by group) (all hormones).
2. perform the Levene's Test for Homogeneity of Variance and Shapiro-Wilk test
3. Q-Q Plots and histograms (all hormones).
4. perform the Brunner-Munzel test, t.test, wilcox.test for 2 independent groups (enter the p-value into the data description table) and repeating the process for multiple variables (all hormones). Conclusion: Which test is applicable?
5. Construct a correlation heatmap for all hormones separately in each of the two groups, selecting the correlation method (Pearson or Spearman) based on the normality of the data, and interpret the resulting patterns.